Randomized trials of endovascular therapy for stroke — impact on stroke care

Maxim Mokin[1], Haydy Rojas[2] and Elad I. Levy[3]

Abstract | Five trials that investigated the efficacy of modern endovascular therapies for stroke — MR CLEAN, ESCAPE, SWIFT PRIME, EXTEND IA and REVASCAT — have been published within the past year, changing the landscape of acute stroke management. The trials used a variety of imaging modalities and combinations of treatment approaches, including the mandatory use of intravenous thrombolysis before the initiation of endovascular therapy. All five trials provided strong evidence to support the use of thrombectomy that is initiated within 6 h of stroke onset, prompting worldwide changes in the guidelines for management of acute stroke by endovascular treatment. The benefits of endovascular therapy were observed irrespective of a patient's age, their NIH Stroke Scale score, or whether they received intravenous thrombolysis. In this article, we review the main findings of these recent trials, focusing on key aspects of their designs, and discuss their impact on the future management of patients with acute stroke that results from large-vessel occlusion. We discuss the values of noncontrast CT ASPECTS, perfusion imaging and angiography for selecting patients to receive endovascular interventions. We also consider the role of thrombectomy beyond 6 h after stroke onset, and in patients with posterior circulation strokes.

[1]*Department of Neurosurgery and Neurology, University of South Florida, 2 Tampa General Circle, 7th Floor, Tampa, Florida 33606, USA.*
[2]*Cardiovascular Services, Tampa General Hospital, 1 Tampa General Circle, Tampa, Florida 33606, USA.*
[3]*Department of Neurosurgery, University at Buffalo, State University of New York, 100 High Street, Suite B4, Buffalo, New York 14226, USA.*
Correspondence to M.M.
maximmokin@gmail.com

doi:10.1038/nrneurol.2015.240
Published online 18 Jan 2016

The clinical benefits and safety of thrombolysis with intravenous recombinant tissue plasminogen activator (IV rtPA) were confirmed in randomized trials, and IV rtPA is the recommended treatment for acute ischaemic stroke in many countries[1,2]. Nevertheless, it has several limitations, including limited efficacy in strokes with a large clot burden, a narrow therapeutic time window, and an increased risk of systemic haemorrhage[3–6]. Endovascular therapies for the treatment of acute ischaemic stroke were developed to overcome these limitations. Until recently, direct evidence in support of catheter-based intra-arterial interventions for stroke was lacking. Within the past year, however, an unprecedented number of trials that tested modern stroke thrombectomy devices were published, thereby changing the landscape of acute stroke management.

The new trials tested a variety of imaging modalities and combinations of treatment approaches, including the mandatory use of intravenous rtPA before initiation of endovascular therapy. Remarkably, in all of these trials, endovascular therapy reduced poststroke disability. In this article, we review the main findings of these recent trials, focusing on key aspects of the trial designs, and discuss their likely impact on the future management of patients with acute stroke that results from large-vessel occlusion.

Evolution of endovascular therapies

Since the publication of the first randomized endovascular stroke trial, PROACT II, which tested intra-arterial recombinant pro-urokinase for the treatment of acute middle cerebral artery (MCA) stroke[7], various revascularization techniques and devices have been tested for the treatment of large-vessel occlusion (BOX 1). Mechanical thrombectomy with the Merci Retriever (a corkscrew-like retrieval device) and aspiration thrombectomy with mechanical clot breakdown provided higher recanalization rates than local pharmacological thrombolysis[8,9]. However, three separate randomized trials (IMS III[10], SYNTHESIS Expansion[11] and MR RESCUE[12]), which compared clinical outcomes after these therapies with either systemic thrombolysis using rtPA or standard medial therapy with antiplatelet agents, found that neither type of thrombectomy reduced poststroke disability. Publication of these three trials in 2013 brought into question the current and future role of endovascular stroke interventions.

At approximately the same time as these trials, a new type of intra-arterial device, called a retrievable stent or stent retriever, was increasing in popularity among neurointerventionalists. Retrievable stents are temporarily deployed at the site of vessel occlusion; once the thrombus is engaged with the stent, both are pulled back through a

© 2016 Macmillan Publishers Limited. All rights reserved

Key points

- Five randomized trials of endovascular stroke therapy published in 2015 provided strong evidence in support of using stent retriever thrombectomy for anterior circulation strokes when initiated within 6 h of stroke onset
- Various imaging approaches are used to select patients to receive endovascular interventions; the most effective imaging modality remains to be determined
- The status of thrombectomy for stroke has now changed from 'experimental' to 'recommended' therapy
- Questions remain about the management of posterior circulation strokes, the time window of effective endovascular treatment beyond 6 h of stroke onset, and the choice of anaesthesia for endovascular procedures
- Future trials are likely to compare thrombectomy devices to optimize the procedure
- Health-care systems will need to adapt to using endovascular therapies, and access to the treatment in remote areas and developing countries will pose a challenge

guide catheter to remove the clot from the body (FIG. 1). Unlike permanent stent deployment, which is often used in percutaneous coronary interventions, use of retrievable stents does not necessitate a long-term commitment to dual antiplatelet therapy, which could increase the risk of intracranial haemorrhage during acute stroke.

Two trials published in 2012 — SWIFT[13] and TREVO2 (REF. 14) — proved that retrievable stents were superior to early mechanical thrombectomy devices: they resulted in higher recanalization rates and more-favourable clinical outcomes than the early-generation mechanical thrombectomy devices. In the USA, where a rigorous FDA approval process is required for medical devices, two brands of stent retrievers are currently approved for use in patients with acute ischaemic stroke (Solitaire™ from Medtronic, and Trevo® from Stryker). In Europe, where the Conformité Européenne device-approval process is more straightforward, multiple types of stent retrievers and adjunct devices that facilitate thrombus extraction are currently approved.

Box 1 | **Intra-arterial approaches to treating acute ischaemic stroke**

Intra-arterial thrombolysis

- The first endovascular treatment modality was launched in the 1990s; PROACT II was the landmark trial[7]
- The pharmacological thrombolytic is delivered locally through a microcatheter
- Very limited ability to achieve good recanalization
- Sometimes used as adjunct therapy for treatment of distal emboli during thrombectomy

Mechanical thrombectomy with the Merci Retriever

- The first mechanical thrombectomy approach for stroke, tested in MERCI and multi-MERCI trials[8,96]
- No longer used in clinical practice following the introduction of stent retrievers

Intracranial stenting

- Emergency intracranial stenting for acute stroke has not been evaluated in randomized trials, only in single-arm studies[97,98]
- Can be used as rescue therapy when thrombectomy is unsuccessful, as often occurs in acute stroke resulting from underlying intracranial atherosclerosis

Stent retriever thrombectomy

- Currently the most rigorously tested and widely used approach to endovascular therapy
- Superiority of stent retrievers over mechanical thrombectomy with the Merci Retriever was demonstrated in the SWIFT and TREVO2 randomized trials[13,14]

Primary aspiration thrombectomy

- Works by placing a catheter immediately next to a clot and applying negative pressure to aspirate the clot
- Used at some institutions before stent retriever thrombectomy[77,99]
- Data on the efficacy of this approach compared with primary stent retriever thrombectomy are limited

New trials of endovascular therapy

Five new and highly successful randomized controlled trials of endovascular therapy, which built on lessons from earlier trials, were published in 2015 (TABLE 1). Retrievable stents were the first-line approach to thrombectomy in each trial. The success rate of thrombectomy and the resulting reductions in disability and mortality among patients with large-vessel-occlusion strokes were so great in the MR CLEAN trial[15], the first of the five to be published, that the other ongoing trials (SWIFT PRIME[16], ESCAPE[17], EXTEND IA[18] and REVASCAT[19]) were halted for interim analysis. Despite population sizes that were smaller than planned, these four trials confirmed the benefit of stent retriever thrombectomy as an adjunct treatment to IV rtPA or as a sole treatment (for patients who were ineligible for intravenous thrombolysis). These five trials provided strong evidence in support of endovascular therapy, regardless of the patient's age, the severity of stroke or the location of the occlusion.

At the time of writing, the results of two additional trials that were stopped early — THRACE[20] and THERAPY[21] — are expected. These trials are excluded from our Review because the final data have not been published. According to preliminary results, however, the THRACE trial revealed improvement of stroke symptoms in patients who were treated with thrombectomy and IV rtPA, compared with IV rtPA alone[22]. The THERAPY trial differed from the other six new trials mentioned, in that it assessed a technique of aspiration thrombectomy. Nevertheless, the reported results show a trend towards better outcomes with aspiration thrombectomy than with IV rtPA, although the trend did not reach statistical significance[23].

Enrolment of patients

The speed at which patients were enrolled for the MR CLEAN trial, which was conducted in the Netherlands, was extremely high: within 3.5 years, 500 patients were enrolled. REVASCAT, another Europe-based trial, had a similarly high enrolment rate, but enrolment was much slower in most other trials of endovascular therapy for stroke, which were conducted in North America and Australia over the same time period.

Several factors explain the differences in enrolment. The high recruitment rate for the MR CLEAN trial resulted from the fact that all stroke centres in the Netherlands participated in the trial and, from 2013, patient costs were reimbursed by insurance companies only if people took part in the trial. REVASCAT was conducted in conjunction with the Catalan government-initiated population-based stroke registry, SONIIA. Participation in SONIIA was mandated by the

© 2016 Macmillan Publishers Limited. All rights reserved

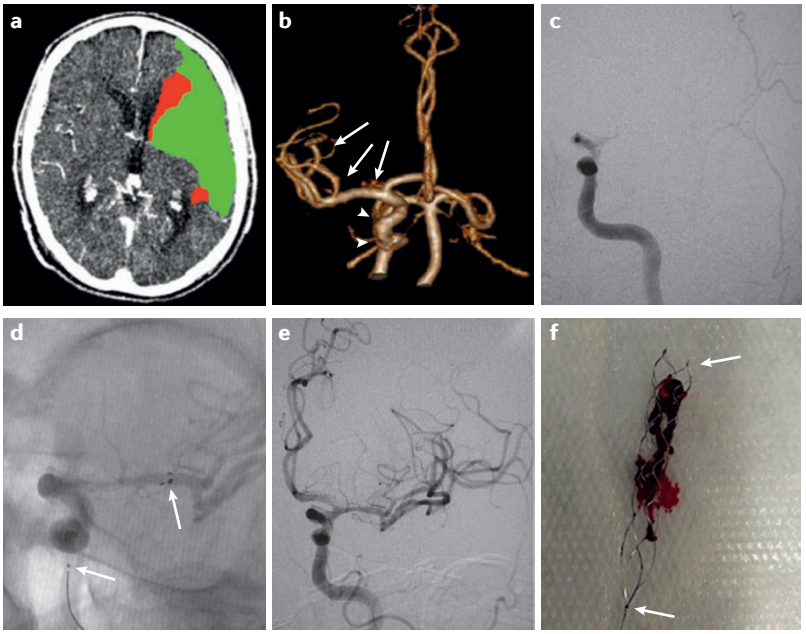

Figure 1 | **Management of acute ischaemic stroke after ICA-T occlusion.** Intravenous thrombolysis was initiated before angiography. **a** | CT perfusion shows a large ischaemic penumbra in the left MCA territory. Red indicates a completed infarct, green indicates tissue that could be salvaged with rapid recanalization and reperfusion. **b** | CT angiography 3D reconstruction shows lack of contrast flow through the left MCA and left ICA intracranially, consistent with occlusion of the left ICA-T. Arrows and arrowheads indicate the patent right MCA and ICA, respectively. **c** | Cerebral angiography confirms persistent occlusion of the left ICA-T. **d** | A stent retriever is deployed within the MCA and ICA-T. Arrows indicate the radiopaque proximal and distal ends of the stent; its body is radiolucent. The stent causes apposition of the clot against the vessel wall and rapidly re-establishes blood flow through the previously occluded vessel segment; a temporary endovascular bypass. **e** | Withdrawal of the stent retriever completely removes the clot, resulting in recanalization of the left ACA and MCA. **f** | Stent retriever with a large clot inside. Arrows indicate the radiopaque markers shown in part **d**. ACA, anterior cerebral artery; ICA, internal carotid artery; ICA-T, ICA terminus; MCA, middle cerebral artery.

government but, unlike in the MR CLEAN trial, strokes could also be treated with endovascular thrombectomy therapy outside REVASCAT if they failed to meet the inclusion criteria[24]. All four major comprehensive stroke centres that treated stroke in Catalonia participated in REVASCAT, thus, every eligible patient was enrolled.

The inclusion criteria for MR CLEAN and REVASCAT were broad, and patients who would have been excluded from the other trials were included. For example, the MR CLEAN trial included patients with anterior cerebral artery occlusion strokes and patients with cervical occlusion or severe stenosis requiring carotid stenting. Enrolment rates were slower in other trials not only as a result of their more rigorous inclusion criteria, but also because, in contrast to the participation policies in the MR CLEAN and REVASCAT trials, many potentially eligible patients were offered interventions without having to participate in trials[25].

Finally, unlike other trials, MR CLEAN did not mandate the use of advanced perfusion imaging or assessment of collateral vessels (discussed in more detail below), and relied only on the use of noncontrast CT Alberta Stroke Program Early CT Score (ASPECTS) to determine the extent of ischaemic damage[26]. By avoiding additional tests, the number of exclusion criteria was minimized, thereby allowing more participants to enter the trial.

Patient selection for endovascular therapy

The switch from the use of early thrombectomy devices to widespread use of stent retrievers was certainly a critical step in the evolution of endovascular therapies, but a better understanding of clinical and imaging criteria for patient selection for thrombectomy is equally important. The NIH Stroke Scale (NIHSS) is one clinical measure used for patient selection. Historically, endovascular therapy was reserved for patients with the most severe neurological deficits, often those with NIHSS scores over 8–10 points. In the 2015 trials of endovascular therapy, median NIHSS scores were 13–18 points. However, patients with low NIHSS scores were eligible for inclusion in the MR CLEAN trial (NIHSS scores ≥2) and the EXTEND IA trial (no lower threshold for NIHSS score).

Imaging is another important tool for patient selection, but the most useful imaging modality remains to be determined. The minimum imaging protocol required to identify patients who are eligible for thrombectomy must include noncontrast CT, to rule out intracranial haemorrhage or major ischaemic injury, and CT angiography (CTA) or magnetic resonance angiography (MRA), to confirm large-vessel occlusion.

CT ASPECTS versus perfusion imaging. Determination of ASPECTS via noncontrast CT is by far the most expedient way to visualize the extent of early ischaemic injury. This score served as the primary imaging criterion for patient selection in the MR CLEAN, ESCAPE and REVASCAT trials. The ASPECTS can be calculated rapidly, allowing efficient exclusion of patients with a burden of ischaemic damage above a certain level (ASPECTS <6 according to current American Heart Association (AHA)–American Stroke Association guidelines), and can be used to predict final infarct size and clinical outcomes; a greater overall score indicates a higher likelihood of neurological improvement as a result of the endovascular intervention[27,28]. The ASPECTS was originally designed to assess the extent of injury in the early stages of stroke in the MCA territory[26]. The score can also be applied to posterior circulation strokes (when it is called pc-ASPECTS), although the limitations of noncontrast CT for imaging posterior fossa structures mean that CTA source images are more valuable in this context, and pc-ASPECTS needs to be studied further[29,30].

In current practice, many stroke centres rely on perfusion imaging to visualize the extent of ischaemic changes and to quantify the brain tissue that might be salvageable (the ischaemic penumbra; FIG. 1). CT-based or MRI-based perfusion imaging is often coupled with CTA or MRA to detect any large-vessel occlusion. Advocates of perfusion imaging argue that noncontrast CT ASPECTS alone is less sensitive than perfusion imaging, and provides less-accurate estimates of the extent of ischaemic changes[31]. A direct comparison of the techniques demonstrated that CT ASPECTS detected regional MCA infarcts with an accuracy of 80% when

© 2016 Macmillan Publishers Limited. All rights reserved

Table 1 | **The 2015 randomized trials of endovascular therapy for stroke**

Trial	Study description	Criteria for patient selection	Treatment groups	Patients achieving functional independence (mRS score 0–2) at 90 days	Number needed to treat for benefit (mRS score 0–2)
MR CLEAN[15]	16 centres in the Netherlands, 500 patients total Enrolment period December 2010–March 2014	CT ASPECTS IA therapy initiated within 6 h of stroke onset	Endovascular arm: stent retrievers were used in 97% of all IA interventions, 87% of IA-treated patients received IV rtPA first Control arm: standard medical therapy (9% of patients) or IV rtPA (91% of patients)	Endovascular group: 33% Control group: 19% OR 1.7 (95% CI 1.2–2.3)	7.1
ESCAPE[17]	22 centres in Canada, USA, South Korea, Ireland and UK, 315 patients total Enrolment period February 2013–October 2014	CT ASPECTS, multiphase CTA to evaluate collateral vessels IA therapy initiated within 12 h of stroke onset	Endovascular arm: stent retrievers were used in 86% of all IA interventions 73% of IA-treated patients received IV rtPA first Control arm: standard medical therapy (21% of patients) or IV rtPA (79% of patients)	Endovascular group: 53% Control group: 29% OR 1.8 (95% CI 1.4–2.4)	4.2
EXTEND IA[18]	Nine centres in Australia and one in New Zealand, 70 patients total Enrolment period August 2012–October 2014	CTP IA therapy initiated within 6 h of stroke onset	Endovascular arm: all patients received IV rtPA plus Solitaire™ stent retriever thrombectomy Control arm: IV rtPA alone	Endovascular group: 71% Control group: 40% OR 4.2 (95% CI 1.4–12)	3.2
SWIFT PRIME[16]	39 centres in USA and Europe, 196 patients total Enrolment period December 2012–November 2014	CTP or MRP, ASPECTS without use of CTP or MRP later in the trial IA therapy initiated within 6 h of stroke onset	Endovascular arm: all patients received IV rtPA plus Solitaire™ stent retriever thrombectomy Control arm: IV rtPA alone	Endovascular group: 60% Control group: 35% OR 1.7 (95% CI 1.2–2.3)	4.0
REVASCAT[19]	Four centres in Spain, 206 patients total Enrolment period November 2012–December 2014	CT ASPECTS score 7–10 or MRI DWI ASPECTS score 6–10 IA therapy initiated within 8 h of stroke onset	Endovascular arm: Solitaire™ stent retriever thrombectomy. 68% of IA-treated patients received IV rtPA first Control arm: standard medical therapy (22% of patients) or IV rtPA (78% of patients)	Endovascular group: 44% Control group: 28% OR 2.1 (95% CI 1.1–4.0)	6.3

ASPECTS, Alberta Stroke Program Early CT Score; CTA, CT angiography; CTP, CT perfusion; DWI, diffusion-weighted imaging; IA, intra-arterial; IV, intravenous; MRP, magnetic resonance perfusion; mRS, modified Rankin scale; rtPA, recombinant tissue plasminogen activator.

used within 3 h of stroke onset, whereas the accuracy of CT perfusion imaging was 97%[32]. In another study that evaluated interobserver agreement in the use of various imaging modalities, the highest agreement between observers was achieved with CT perfusion imaging, followed by CTA source imaging, then noncontrast CT ASPECTS (correlation coefficients 0.903, 0.876 and 0.834, respectively)[33]. Perfusion-guided imaging, therefore, shows promise as a tool for selecting patients to receive endovascular therapy according to the size of their infarct rather than the time of symptom onset, which is the basis for time-guided selection[34].

Perfusion imaging was used in the selection of patients for the EXTEND-IA and SWIFT PRIME trials. Only CT perfusion imaging was used to evaluate potential participants of the EXTEND-IA trial, whereas CT and MRI perfusion imaging were used in the SWIFT PRIME trial, although CT perfusion alone was used in the majority of cases (80%)[35]. In both trials, functional independence of patients who received endovascular therapy was markedly greater than that in the control groups (absolute difference 31% in EXTEND-IA and 25% in SWIFT PRIME).

One concern with the use of perfusion imaging is that the imaging and processing requires more time than does CT ASPECTS, and can, therefore, prolong the time between arrival at hospital and initiation of endovascular therapy. In one retrospective study conducted in 10 stroke centres, the time from imaging to initiation of endovascular therapy (groin puncture) was 53 min longer when CT perfusion imaging and noncontrast CT imaging were used, as opposed to noncontrast CT

© 2016 Macmillan Publishers Limited. All rights reserved

imaging alone[36]. Clinical outcomes and complication rates were similar in both groups, bringing into question the value of perfusion imaging for stroke.

CT and MR angiography. Use of CTA or MRA has become a commonly used standard test for confirming large-vessel occlusion. All five of the 2015 trials of endovascular therapy required the use of noninvasive angiography to confirm the presence of a target occlusion before random allocation of participants to treatment groups. This requirement was a fundamental difference from the earlier IMS III and SYNTHESIS Expansion trials, in which some patients who did not have a target occlusion were included.

The definition of large-vessel occlusion varied between the five trials. Patients with strokes that involved the intracranial internal carotid artery or the MCA M1 segment were eligible for inclusion in all five trials. Additional patients that were included in some trials were those with: occlusion of the MCA M2 segment (ESCAPE, MR CLEAN and EXTEND IA), occlusion of the proximal anterior cerebral artery (MR CLEAN), and extracranial carotid artery dissection, stenosis or acute occlusion (MR CLEAN and REVASCAT).

CTA also enables assessment of intracranial collateral vessels (collateral vessels at the occlusion site can be compared with unaffected vessels on the contralateral side), which is a powerful predictor of clinical outcomes after revascularization; its value in this respect is similar to that of perfusion imaging. Well-developed collateral vessels lead to smaller final infarct volumes and higher rates of functional independence[37–40]. Even more-precise visualization of collateral vessels is possible with multiphase CTA, in which cranial scans are repeated in the early and late venous phases without the use of a contrast agent. As use of multiphase CTA rather than CT perfusion imaging eliminates the need for a contrast agent and a separate scan, this approach reduces the time until initiation of therapy, although this test involves a higher dose of radiation[41,42]. In the ESCAPE trial, multiphase CTA was the main imaging modality used to assess collateral vessel status; only patients with good or moderate collateral vessels (≥50% of the affected territory of the pial arterial circulation undamaged) were included.

Outcomes

Number needed to treat. The number needed to treat (NNT) is commonly used as a surrogate of treatment efficacy in clinical trials. The measure indicates the number of patients to which a treatment must be administered before either a single patient achieves a benefit or one adverse event is prevented. The NNT outcomes in the 2015 trials of endovascular therapy were remarkably low: as few as 3.2 patients in the EXTEND IA trial and 4.0 patients in the SWIFT PRIME trial (based on a dichotomy of favourable versus poor outcome at 3 months after treatment; TABLE 1). This result proves that thrombectomy has a very strong treatment effect. For comparison, in a general population and when all stroke types are included, a similar NNT of 3.6 can only be achieved with intravenous thrombolysis in patients who are treated within 90 min of stroke onset[43]. If rtPA is administered within 3.0–4.5 h, the NNT rises to 5.9 patients, and beyond the 4.5 h window, the NNT is 19.3 patients. However, when the fact that strokes resulting from a large-vessel occlusion are more resistant to intravenous thrombolysis than are other types of strokes is taken into account, the odds ratios for good neurological outcome are higher in the recent thrombectomy trials than in intravenous thrombolysis trials, even when only treatment within the first 90 min after stroke onset is considered. When making indirect comparisons of outcomes between the five trials, such as comparing modified Rankin scale scores with NNT values, differences in imaging selection criteria, sample sizes and treatment protocols should be considered as potential sources of bias.

Time to treatment. The data from the 2015 trials of endovascular therapy provide strong evidence in support of thrombectomy that is initiated within 6 h of stroke onset. Enrolment for the ESCAPE trial was permitted up to 12 h after stroke onset, but the median time to treatment initiation was only 169 min, and reperfusion was achieved within 241 min. Only 49 participants (16%) received treatment beyond the 6 h window but, even in this small population, a strong trend in favour of thrombectomy was evident when the rates of long-term functional independence were compared with those of the control group (medical therapy only; OR 1.7, 95% CI 0.7–4.0).

IV rtPA and intra-arterial thrombectomy. Results from the 2015 trials of endovascular therapy go some way towards determining whether IV rtPA treatment offers any benefit when administered alongside endovascular therapy. Most patients across the five trials received IV rtPA. In the SWIFT PRIME and EXTEND IA trials, all patients who underwent endovascular therapy received IV rtPA; the MR CLEAN, ESCAPE and REVASCAT trials included patients who were ineligible for IV rtPA, but the majority of patients who were treated with endovascular therapy received IV rtPA first. Only a minority of patients, therefore, were taken straight to the angiography suite, but subgroup analyses revealed a clinical benefit of endovascular interventions without prior use of IV rtPA. Indeed, in the ESCAPE trial, patients who did not receive IV rtPA gained greater functional independence than did patients treated with IV thrombolysis before endovascular therapy was initiated (OR 2.6). Furthermore, in the REVASCAT trial, patients who did not receive IV rtPA before thrombectomy had higher rates of recovery than did patients who received IV rtPA first (OR 2.7 in patients who did not receive IV rtPA, and OR 1.4 in those who received IV rtPA first, versus best medical therapy alone, both in favour of thrombectomy).

Age and stroke severity. Previous studies have indicated that older age and greater severity of stroke symptoms are associated with poorer clinical outcomes after thrombectomy with stent retrievers[44–46]. Prespecified

© 2016 Macmillan Publishers Limited. All rights reserved

subgroups for analysis of the effects of age and stroke severity differed among the 2015 trials of endovascular therapy, but the benefit of thrombectomy was seen irrespective of age or NIHSS scores.

Unanswered questions

The 2015 trials of endovascular therapy have undoubtedly provided valuable insight into the efficacy of thrombectomy in a subset of patients. Nevertheless, several important questions remain unanswered, and further investigation is now needed to address these issues.

Posterior circulation strokes

Patients with posterior circulation strokes were excluded from the 2015 trials of endovascular therapy (except the THRACE trial[47], which included patients with strokes that resulted from distal basilar artery occlusion). However, these types of stroke — especially when caused by occlusion of the basilar artery — are associated with extremely high morbidity and mortality. One meta-analysis of stent retriever thrombectomy for basilar artery occlusion demonstrated that the treatment allowed 42% of patients to achieve functional independence after 3 months[48]. However, mortality was 30%, much higher than in the 2015 trials of endovascular therapy that included only patients with anterior circulation stroke. The ENDSTROKE registry also included 148 patients with basilar artery occlusion, 84% of whom were treated with stent retrievers[49]. 34% of these patients had gained functional independence after 3 months, but 35% died.

Historical data from retrospective studies have shown that medical management or IV rtPA treatment of acute basilar artery occlusion results in poor outcomes[50–53]. The persistence of high mortality, even with modern thrombectomy approaches, demands more-effective treatment protocols for posterior circulation strokes.

Time to reperfusion

Analysis of data from the IMS III trial showed that an increased time to reperfusion was associated with a decreased likelihood of favourable clinical outcomes[54,55]. Analysis of combined data from the SWIFT trial and STAR[56] — a prospective single-arm study of the Solitaire™ stent retriever in acute stroke — confirmed the importance of fast reperfusion for good outcomes[57]: every 5 min delay before endovascular reperfusion worsened the clinical outcome for 1 of 100 patients. Reperfusion within 2 h was associated with a >80% chance of good clinical outcomes, but this chance was reduced to <20% when reperfusion took 10 h.

The 2015 trials of endovascular therapy recognized that minimization of delays in initiation of revascularization procedures is crucial for the success of the treatment. No randomized trials to date have been able to prove the efficacy and safety of endovascular interventions beyond the first 6 h after stroke onset. The ESCAPE trial included a subgroup of patients who were treated within 6–12 h of stroke onset, but the small size of the subgroup meant that statistical power was insufficient to determine whether the treatment was of any benefit. The DAWN trial[58] is currently comparing the effects of stent retriever thrombectomy with those of standard medical therapy when treatment is initiated at 6–24 h after stroke onset. Such comparisons will be helpful in the management of patients who present with so-called wake-up stroke, in which the exact time of stroke onset is unknown and the 'last known well' time is used as a reference instead. Retrospective data suggest that stent retriever thrombectomy in patients with wake-up stroke results in rates of good clinical outcomes that approach those seen in the 2015 trials of endovascular therapy[59,60].

Choice of anaesthesia

Whether conscious sedation or general anaesthesia is used during thrombectomy is often left to the discretion of the neurointerventionalist. The MR CLEAN pretrial study group found that the use of general anaesthesia in patients with anterior circulation stroke was associated with poor clinical outcomes[61]. Several other retrospective studies have shown a similar relationship between the use of general anaesthesia and a higher chance of poor neurological outcomes and mortality[62–64]. The Sedation versus Intubation for Endovascular Stroke Treatment (SIESTA)[65] trial is an ongoing prospective randomized trial that aims to determine whether conscious sedation is the optimal choice for periprocedural management of patients undergoing thrombectomy for stroke.

Future directions

The future of trials

Strong evidence in support of using thrombectomy over other available treatments now raises ethical concerns regarding the assignment of patients with large-vessel occlusion to receive noninterventional therapies in randomized trials. In the field of interventional cardiology, proof of the clinical benefits of percutaneous coronary interventions was rapidly followed by a shift in the designs of clinical trials towards head-to-head comparisons of devices to identify which was best for revascularization of coronary arteries. We should expect to witness similar transformations in trials of endovascular interventions in stroke. Various types of stent retrievers and adjunct tools, such as delivery catheters, are available, and participation of the device-manufacturing industry is likely to increase in future trials. Recanalization rates have dramatically improved with the use of retrievable stents rather than early thrombectomy devices but, even in the 2015 trials, recanalization was unsuccessful in many patients: 34% in the REVASCAT trial and 28% in the ESCAPE trial, for example. These results demonstrate the need to further improve the available technologies.

Increasing success

Successful recanalization and reperfusion is one of the most powerful predictors of clinical success in endovascular stroke interventions[66]. This success is assessed immediately after the thrombectomy procedure by combining digital subtraction angiography and the Thrombolysis in Cerebral Infarction (TICI) scale; a score of 2b or 3 indicates a technical success[67,68]. Nevertheless,

© 2016 Macmillan Publishers Limited. All rights reserved

thrombectomy in stroke does not achieve reperfusion rates that are comparable to those achieved with modern coronary interventions (TICI 2/3 rate range was 59–86% in the 2015 trials, compared with >90% reperfusion rates for many coronary interventions), so technical advances are needed to improve reperfusion rates. Technical tests of currently available stroke thrombectomy devices in bench research and animal models of stroke have produced promising results, but the data collected so far are conflicting, and interpretation is limited by difficulty in replicating and simulating the complexity of intracranial human circulation in the laboratory[69–73].

The optimal approach to thrombectomy — that is, the one that achieves the highest recanalization rates with the shortest procedure duration — has yet to be determined. Evidence suggests that recanalization rates are improved by the use of adjunct devices, such as guide catheters with distal balloons for flow arrest, or by continuous aspiration with a large-bore catheter during thrombectomy with a stent retriever[74,75]. However, patient selection criteria were not reported consistently and incomplete technical details were provided in these studies, thereby limiting interpretation of the data. Direct comparisons of the safety and efficacy of different thrombectomy devices would require randomized trials with designs similar to those of the SWIFT and TREVO2 trials[13,14], which evaluated and compared technical aspects of the two types of thrombectomy device (early-generation Merci corkscrew retriever versus stent retriever). Aspiration thrombectomy with recently developed, new-generation distal aspiration catheters might improve recanalization rates, but indirect comparison of this approach with stent retrievers is subject to bias and controversy due to conflicting results on their efficacy in the literature[76–78].

Clinical use of thrombectomy

In light of the strong evidence for the benefits of endovascular therapy, the status of thrombectomy for stroke has changed from 'experimental' to 'recommended' therapy. Updated guidelines on mechanical thrombectomy that reflect the results of the 2015 trials have been released by the AHA in North America[79]. The guidelines state that class I, level A evidence supports the use of stent retriever thrombectomy that is initiated within 6 h of stroke onset in patients with anterior circulation, large-vessel occlusion and who are receiving intravenous thrombolysis. A preliminary consensus statement released by the European Stroke Organization (ESO)-Karolinska Stroke Update also included a recommendation that mechanical thrombectomy should be used in addition to intravenous thrombolysis within 6 h of stroke onset in patients with large-vessel occlusion, anterior circulation strokes. The ESO guidelines include a wider range of indications for mechanical thrombectomy than did the AHA guidelines, which are more rigid and require a higher level of evidence from randomized trials when providing recommendations for clinical and treatment strategies. The ESO also recommends thrombectomy as the first-line treatment in patients with large-vessel occlusion strokes and contraindications for intravenous thrombolysis. The Society of NeuroInterventional Surgery recently released a similar statement that supports the use of thrombectomy within 6 h of stroke onset[80].

The impact of the 2015 trials of endovascular therapy for stroke is likely to differ considerably between countries. Such differences are already seen with the use of intravenous thrombolysis. In North America, Japan and Western Europe, the proportion of patients treated with intravenous thrombolysis is relatively high, at 1–11%[81,82] (see also the Royal College of Physicians Sentinel Stroke National Audit Programme (SSNAP) April–June 2015 summary report).

Financial constraints and poor infrastructure mean that these numbers are much lower in other regions, such as Eastern Europe, the Middle East and India[83–85]. According to some studies, as many as 22–44% of patients with acute stroke arrive at hospital in time for endovascular therapy to be initiated within 6 h of symptom onset[86–88], but the higher cost of endovascular therapy and the need for specialized imaging to detect large-vessel occlusion means that provision of patient access to endovascular therapy in developing countries will be a challenge.

Furthermore, all hospitals that provide care for patients with acute stroke will need to adapt. Hospitals that offer endovascular interventions will need to ensure that the workflow is optimized to deliver such services without delays. Hospitals that cannot conduct neurointerventional procedures (currently the majority of hospitals) will need to form networks with dedicated stroke centres (preferably comprehensive stroke centres) and establish routes of rapid communication to ensure that patients who are eligible for thrombectomy are transferred to appropriately equipped hospitals. Emergency medical personnel and stroke physicians will need to correctly recognize patients with suspected large-vessel occlusion. Furthermore, expectations of stroke centres that offer thrombectomy will need to be standardized and their performance assessed by medical societies and national committees in an arrangement that is similar to the current requirements for the use of IV rtPA in acute stroke.

Future work should also focus on the increasing roles of emergency medical services, prehospital triage and stroke code protocols in the emergency department in reducing door-to-needle time[89–91]. An example can be taken from the management of patients with ST elevation myocardial infarction, who — throughout North America at least — are now directly transferred to hospitals that are capable of performing percutaneous coronary interventions as standard. When transfers cannot be expedited, the 'ship-and-drip' treatment paradigm is implemented, and intravenous fibrinolytics are started before the patient is transferred to the percutaneous coronary interventions suite. Much like this cardiological paradigm, stroke triage protocols need to be tailored to specific populations and hospital networks, such as densely populated urban regions and rural areas with limited access to endovascular therapy. Telestroke, which allows remote real-time evaluation of patients with stroke

© 2016 Macmillan Publishers Limited. All rights reserved

by a neurologist using videoconferencing and other sources of electronic communication, and other novel approaches, such as the use of mobile stroke units, could help to provide emergency neurointerventional care in remote or rural areas[92–95].

Conclusions

Results from the 2015 trials of endovascular therapy for stroke provide strong evidence in support of thrombectomy initiated within 6 h of stroke onset. Although the majority of patients who were treated with thrombectomy received IV rtPA first, reduced disability also resulted from revascularization alone. These trials primarily tested thrombectomy with retrievable stents, which offer higher recanalization rates than the early-generation thrombectomy devices used in the 2013 trials. Additional research is needed to further improve the efficacy of stent retrievers to achieve reperfusion rates comparable to those achieved with modern coronary interventions. Furthermore, at this point, no high-quality data are available on how primary aspiration thrombectomy with new-generation distal aspiration catheters compares to stent retriever thrombectomy in terms of safety and efficacy.

The 2015 trials differed in the imaging protocols used for patient selection, ranging from noncontrast CT alone to perfusion imaging and multiphase CTA. Owing to this variation, we do not yet know which imaging modality will prove to be of the highest value for the selection of patients to receive endovascular stroke interventions. All five trials required confirmation of large-vessel occlusion via CTA or MRA. Combined with noninvasive CT to rule out haemorrhage and measure ASPECTS, these tests can provide clinicians with the basic information required when evaluating potential candidates for thrombectomy.

1. The National Institute of Neurological Disorders and Stroke rt-PA Stroke Study Group. Tissue plasminogen activator for acute ischemic stroke. *N. Engl. J. Med.* **333**, 1581–1587 (1995).
2. Hacke, W. *et al.* Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke. *N. Engl. J. Med.* **359**, 1317–1329 (2008).
3. Faiz, K. W., Sundseth, A., Thommessen, B. & Ronning, O. M. Reasons for low thrombolysis rate in a Norwegian ischemic stroke population. *Neurol. Sci.* **35**, 1977–1982 (2014).
4. García-Moncó, J. C. *et al.* Analysis of the reasons for exclusion from tPA therapy after early arrival in acute stroke patients. *Clin. Neurol. Neurosurg.* **109**, 50–53 (2007).
5. Barber, P. A., Zhang, J., Demchuk, A. M., Hill, M. D. & Buchan, A. M. Why are stroke patients excluded from TPA therapy? An analysis of patient eligibility. *Neurology* **56**, 1015–1020 (2001).
6. van den Berg, J. S. & de Jong, G. Why ischemic stroke patients do not receive thrombolytic treatment: results from a general hospital. *Acta Neurol. Scand.* **120**, 157–160 (2009).
7. Furlan, A. *et al.* Intra-arterial prourokinase for acute ischemic stroke. The PROACT II study: a randomized controlled trial. *JAMA* **282**, 2003–2011 (1999).
8. Smith, W. S. *et al.* Safety and efficacy of mechanical embolectomy in acute ischemic stroke: results of the MERCI trial. *Stroke* **36**, 1432–1438 (2005).
9. Penumbra Pivotal Stroke Trial Investigators. The Penumbra Pivotal Stroke Trial: safety and effectiveness of a new generation of mechanical devices for clot removal in intracranial large vessel occlusive disease. *Stroke* **40**, 2761–2768 (2009).
10. Broderick, J. P. *et al.* Endovascular therapy after intravenous t-PA versus t-PA alone for stroke. *N. Engl. J. Med.* **368**, 893–903 (2013).
11. Ciccone, A. *et al.* Endovascular treatment for acute ischemic stroke. *N. Engl. J. Med.* **368**, 904–913 (2013).
12. Kidwell, C. S. *et al.* A trial of imaging selection and endovascular treatment for ischemic stroke. *N. Engl. J. Med.* **368**, 914–923 (2013).
13. Saver, J. L. *et al.* Solitaire flow restoration device versus the Merci Retriever in patients with acute ischaemic stroke (SWIFT): a randomised, parallel-group, non-inferiority trial. *Lancet* **380**, 1241–1249 (2012).
14. Nogueira, R. G. *et al.* Trevo versus Merci retrievers for thrombectomy revascularisation of large vessel occlusions in acute ischaemic stroke (TREVO 2): a randomised trial. *Lancet* **380**, 1231–1240 (2012).
15. Berkhemer, O. A. *et al.* A randomized trial of intraarterial treatment for acute ischemic stroke. *N. Engl. J. Med.* **372**, 11–20 (2015).
16. Saver, J. L. *et al.* Stent-retriever thrombectomy after intravenous t-PA versus t-PA alone in stroke. *N. Engl. J. Med.* **372**, 2285–2295 (2015).
17. Goyal, M. *et al.* Randomized assessment of rapid endovascular treatment of ischemic stroke. *N. Engl. J. Med.* **372**, 1019–1030 (2015).
18. Campbell, B. C. *et al.* Endovascular therapy for ischemic stroke with perfusion-imaging selection. *N. Engl. J. Med.* **372**, 1009–1018 (2015).
19. Jovin, T. G. *et al.* Thrombectomy within 8 hours after symptom onset in ischemic stroke. *N. Engl. J. Med.* **372**, 2296–2306 (2015).
20. US National Library of Medicine. *ClinicalTrials.gov* [online], https://clinicaltrials.gov/ct2/show/NCT01062698 (2015).
21. US National Library of Medicine. *ClinicalTrials.gov* [online], https://clinicaltrials.gov/ct2/show/NCT01429350 (2015).
22. Bracard, S., Guillemin, F., Ducrocq, X. & the THRACE investigators. Thrace study: intermediate analysis results [abstract ESOC-1594]. *Int. J. Stroke* **10** (Suppl. 2), 31 (2015).
23. Mocco, J. *et al.* Results of the therapy trial: a prospective, randomized trial to define the role of mechanical thrombectomy as adjunctive treatment to IV rtPA in acute ischemic stroke [abstract ESOC-1614]. *Int. J. Stroke* **10** (Suppl. 2), 10 (2015).
24. Urra, X. *et al.* Mechanical thrombectomy in and outside the REVASCAT trial: insights from a concurrent population-based stroke registry. *Stroke* **46**, 3437–3442 (2015).
25. Goyal, M. *et al.* Endovascular stroke trials: why we must enroll all eligible patients. *Stroke* **44**, 3591–3595 (2013).
26. Barber, P. A., Demchuk, A. M., Zhang, J. & Buchan, A. M. Validity and reliability of a quantitative computed tomography score in predicting outcome of hyperacute stroke before thrombolytic therapy. *Lancet* **355**, 1670–1674 (2000).
27. Hill, M. D. *et al.* Alberta Stroke Program early computed tomography score to select patients for endovascular treatment: Interventional Management of Stroke (IMS)-III Trial. *Stroke* **45**, 444–449 (2014).
28. Yoo, A. J. *et al.* Impact of pretreatment noncontrast CT Alberta Stroke Program Early CT Score on clinical outcome after intra-arterial stroke therapy. *Stroke* **45**, 746–751 (2014).
29. Puetz, V. *et al.* Extent of hypoattenuation on CT angiography source images in basilar artery occlusion: prognostic value in the Basilar Artery International Cooperation Study. *Stroke* **42**, 3454–3459 (2011).
30. Puetz, V. *et al.* Extent of hypoattenuation on CT angiography source images predicts functional outcome in patients with basilar artery occlusion. *Stroke* **39**, 2485–2490 (2008).
31. Lum, C. *et al.* Computed tomographic angiography and cerebral blood volume can predict final infarct volume and outcome after recanalization. *Stroke* **45**, 2683–2688 (2014).
32. Lin, K. *et al.* Accuracy of the Alberta Stroke Program Early CT Score during the first 3 hours of middle cerebral artery stroke: comparison of noncontrast CT, CT angiography source images, and CT perfusion. *AJNR Am. J. Neuroradiol.* **29**, 931–936 (2008).
33. Finlayson, O. *et al.* Interobserver agreement of ASPECT score distribution for noncontrast CT, CT angiography, and CT perfusion in acute stroke. *Stroke* **44**, 234–236 (2013).
34. Turk, A. S. *et al.* CT perfusion-guided patient selection for endovascular recanalization in acute ischemic stroke: a multicenter study. *J. Neurointerv. Surg.* **5**, 523–527 (2013).
35. Albers, G. W. *et al.* Relationships between imaging assessments and outcomes in solitaire with the intention for thrombectomy as primary endovascular treatment for acute ischemic stroke. *Stroke* **46**, 2786–2794 (2015).
36. Sheth, K. N. *et al.* Advanced modality imaging evaluation in acute ischemic stroke may lead to delayed endovascular reperfusion therapy without improvement in clinical outcomes. *J. Neurointerv. Surg.* **5** (Suppl. 1), i62–i65 (2013).
37. Liebeskind, D. S. *et al.* Impact of collaterals on successful revascularization in Solitaire FR with the intention for thrombectomy. *Stroke* **45**, 2036–2040 (2014).
38. Elijovich, L. *et al.* CTA collateral score predicts infarct volume and clinical outcome after endovascular therapy for acute ischemic stroke: a retrospective chart review. *J. Neurointerv. Surg.* http://dx.doi.org/10.1136/neurintsurg-2015-011731.
39. Sung, S. M. *et al.* Functional outcome after recanalization for acute pure M1 occlusion of the middle cerebral artery as assessed by collateral CTA flow. *Clin. Neurol. Neurosurg.* **131**, 72–76 (2015).
40. Nambiar, V. *et al.* CTA collateral status and response to recanalization in patients with acute ischemic stroke. *AJNR Am. J. Neuroradiol.* **35**, 884–890 (2014).
41. Yang, C. Y. *et al.* Multiphase CT angiography versus single-phase CT angiography: comparison of image quality and radiation dose. *AJNR Am. J. Neuroradiol.* **29**, 1288–1295 (2008).
42. Menon, B. K. *et al.* Multiphase CT angiography: a new tool for the imaging triage of patients with acute ischemic stroke. *Radiology* **275**, 510–520 (2015).
43. Lansberg, M. G., Schrooten, M., Bluhmki, E., Thijs, V. N. & Saver, J. L. Treatment time-specific number needed to treat estimates for tissue plasminogen activator therapy in acute stroke based on shifts over the entire range of the modified Rankin Scale. *Stroke* **40**, 2079–2084 (2009).
44. Daou, B. *et al.* Predictors of outcome, complications, and recanalization of the Solitaire device: a study of 89 cases. *Neurosurgery* **77**, 355–361 (2015).
45. Mokin, M. *et al.* Solitaire Flow Restoration thrombectomy for acute ischemic stroke: retrospective multicenter analysis of early postmarket experience after FDA approval. *Neurosurgery* **73**, 19–25 (2013).
46. Raoult, H. *et al.* Prognostic factors for outcomes after mechanical thrombectomy with solitaire stent. *J. Neuroradiol.* **40**, 252–259 (2013).
47. Khatri, P. *et al.* State of acute endovascular therapy: report from the 12th Thrombolysis, Thrombectomy, and Acute Stroke Therapy Conference. *Stroke* **46**, 1727–1734 (2015).

© 2016 Macmillan Publishers Limited. All rights reserved

48. Gory, B. *et al.* Outcomes of stent retriever thrombectomy in basilar artery occlusion: an observational study and systematic review. *J. Neurol. Neurosurg. Psychiatry* http://dx.doi.org/10.1136/jnnp-2014-310250.
49. Singer, O. C. *et al.* Mechanical recanalization in basilar artery occlusion: the ENDOSTROKE study. *Ann. Neurol.* **77**, 415–424 (2015).
50. Schonewille, W. J., Algra, A., Serena, J., Molina, C. A. & Kappelle, L. J. Outcome in patients with basilar artery occlusion treated conventionally. *J. Neurol. Neurosurg. Psychiatry* **76**, 1238–1241 (2005).
51. Baird, T. A., Muir, K. W. & Bone, I. Basilar artery occlusion. *Neurocrit. Care* **1**, 319–329 (2004).
52. Lindsberg, P. J. & Mattle, H. P. Therapy of basilar artery occlusion: a systematic analysis comparing intra-arterial and intravenous thrombolysis. *Stroke* **37**, 922–928 (2006).
53. Sairanen, T. *et al.* Intravenous thrombolysis of basilar artery occlusion: predictors of recanalization and outcome. *Stroke* **42**, 2175–2179 (2011).
54. Khatri, P. *et al.* Time to angiographic reperfusion and clinical outcome after acute ischaemic stroke: an analysis of data from the Interventional Management of Stroke (IMS III) Phase 3 trial. *Lancet Neurol.* **13**, 567–574 (2014).
55. Goyal, M. *et al.* Evaluation of interval times from onset to reperfusion in patients undergoing endovascular therapy in the Interventional Management of Stroke III trial. *Circulation* **130**, 265–272 (2014).
56. Pereira, V. M. *et al.* Prospective, multicenter, single-arm study of mechanical thrombectomy using Solitaire Flow Restoration in acute ischemic stroke. *Stroke* **44**, 2802–2807 (2013).
57. Sheth, S. A. *et al.* Time to endovascular reperfusion and degree of disability in acute stroke. *Ann. Neurol.* **78**, 584–593 (2015).
58. US National Library of Medicine. *ClinicalTrials.gov* [online], https://clinicaltrials.gov/ct2/show/NCT02142283 (2015).
59. Mokin, M. *et al.* Endovascular therapy of wake-up strokes in the modern era of stent retriever thrombectomy. *J. Neurointerv. Surg.* http://dx.doi.org/10.1136/neurintsurg-2014-011586.
60. Buck, D., Shaw, L. C., Price, C. I. & Ford, G. A. Reperfusion therapies for wake-up stroke: systematic review. *Stroke* **45**, 1869–1875 (2014).
61. van den Berg, L. A. *et al.* Type of anesthesia and differences in clinical outcome after intra-arterial treatment for ischemic stroke. *Stroke* **46**, 1257–1262 (2015).
62. Abou-Chebl, A. *et al.* Conscious sedation versus general anesthesia during endovascular therapy for acute anterior circulation stroke: preliminary results from a retrospective, multicenter study. *Stroke* **41**, 1175–1179 (2010).
63. John, S. *et al.* Intra-arterial therapy for acute ischemic stroke under general anesthesia versus monitored anesthesia care. *Cerebrovasc. Dis.* **38**, 262–267 (2014).
64. Brinjikji, W. *et al.* Conscious sedation versus general anesthesia during endovascular acute ischemic stroke treatment: a systematic review and meta-analysis. *AJNR Am. J. Neuroradiol.* **36**, 525–529 (2015).
65. US National Library of Medicine. *ClinicalTrials.gov* [online], https://clinicaltrials.gov/ct2/show/NCT02126085?term = NCT02126085&rank = 1 (2015).
66. Rha, J. H. & Saver, J. L. The impact of recanalization on ischemic stroke outcome: a meta-analysis. *Stroke* **38**, 967–973 (2007).
67. Zaidat, O. O. *et al.* Recommendations on angiographic revascularization grading standards for acute ischemic stroke: a consensus statement. *Stroke* **44**, 2650–2663 (2013).
68. Marks, M. P. *et al.* Correlation of AOL recanalization, TIMI reperfusion and TICI reperfusion with infarct growth and clinical outcome. *J. Neurointerv. Surg.* **6**, 724–728 (2014).
69. Gounis, M. J., Wakhloo, A. K. & Chueh, J. Y. Preclinical investigations for thrombectomy devices — does it translate to humans? *Stroke* **44**, S7–S10 (2013).
70. Teng, D. *et al.* Endothelial trauma from mechanical thrombectomy in acute stroke: *in vitro* live-cell platform with animal validation. *Stroke* **46**, 1099–1106 (2015).
71. Chueh, J. Y., Puri, A. S., Wakhloo, A. K. & Gounis, M. J. Risk of distal embolization with stent retriever thrombectomy and ADAPT. *J. Neurointerv. Surg.* http://dx.doi.org/10.1136/neurintsurg-2014-011491.
72. Mokin, M., Setlur Nagesh, S. V., Ionita, C. N., Levy, E. I. & Siddiqui, A. H. Comparison of modern stroke thrombectomy approaches using an *in vitro* cerebrovascular occlusion model. *AJNR Am. J. Neuroradiol.* **36**, 547–551 (2015).
73. Chueh, J. Y. *et al.* Reduction in distal emboli with proximal flow control during mechanical thrombectomy: a quantitative *in vitro* study. *Stroke* **44**, 1396–1401 (2013).
74. Nguyen, T. N. *et al.* Balloon guide catheter improves revascularization and clinical outcomes with the Solitaire device: analysis of the North American Solitaire Acute Stroke Registry. *Stroke* **45**, 141–145 (2014).
75. Humphries, W. *et al.* Distal aspiration with retrievable stent assisted thrombectomy for the treatment of acute ischemic stroke. *J. Neurointerv. Surg.* **7**, 90–94 (2015).
76. Mehta, B. P. *et al.* ADAPT FAST Study: third-generation stroke thrombectomy devices place renewed focus on the elusive relationship between revascularization and good outcomes. *J. Neurointerv. Surg.* http://dx.doi.org/10.1136/neurintsurg-2014-011507.
77. Turk, A. S. *et al.* ADAPT FAST study: a direct aspiration first pass technique for acute stroke thrombectomy. *J. Neurointerv. Surg.* **6**, 260–264 (2014).
78. Fiorella, D. *et al.* Too much guidance. *J. Neurointerv. Surg.* **7**, 626–627 (2015).
79. Powers, W. J. *et al.* 2015 American Heart Association/American Stroke Association focused update of the 2013 guidelines for the early management of patients with acute ischemic stroke regarding endovascular treatment: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* **46**, 3020–3035 (2015).
80. Jayaraman, M. V. *et al.* Embolectomy for stroke with emergent large vessel occlusion (ELVO): report of the Standards and Guidelines Committee of the Society of NeuroInterventional Surgery. *J. Neurointerv. Surg.* **7**, 316–321 (2015).
81. Schwamm, L. H. *et al.* Temporal trends in patient characteristics and treatment with intravenous thrombolysis among acute ischemic stroke patients at Get With The Guidelines-Stroke hospitals. *Circ. Cardiovasc. Qual. Outcomes* **6**, 543–549 (2013).
82. Wang, Y. *et al.* Using recombinant tissue plasminogen activator to treat acute ischemic stroke in China: analysis of the results from the Chinese National Stroke Registry (CNSR). *Stroke* **42**, 1658–1664 (2011).
83. Ghandehari, K. Barriers of thrombolysis therapy in developing countries. *Stroke Res. Treat.* **2011**, 686797 (2011).
84. Lenti, L. *et al.* Stroke care in Central Eastern Europe: current problems and call for action. *Int. J. Stroke* **8**, 365–371 (2013).
85. Wasay, M., Khatri, I. A. & Kaul, S. Stroke in South Asian countries. *Nat. Rev. Neurol.* **10**, 135–143 (2014).
86. Song, D. *et al.* Factors associated with early hospital arrival in patients with acute ischemic stroke. *J. Stroke* **17**, 159–167 (2015).
87. Leon-Jimenez, C. *et al.* Hospital arrival time and functional outcome after acute ischaemic stroke: results from the PREMIER study. *Neurologia* **29**, 200–209 (2014).
88. Sauser, K., Bravata, D. M., Hayward, R. A. & Levine, D. A. A national evaluation of door-to-imaging times among acute ischemic stroke patients within the Veterans Health Administration. *J. Stroke Cerebrovasc. Dis.* **24**, 1329–1332 (2015).
89. Broderick, J. P. *et al.* Evolution of practice during the Interventional Management of Stroke III Trial and implications for ongoing trials. *Stroke* **45**, 3606–3611 (2014).
90. Busby, L. *et al.* CODE FAST: a quality improvement initiative to reduce door-to-needle times. *J. Neurointerv. Surg.* http://dx.doi.org/10.1136/neurintsurg-2015-011806.
91. Liebeskind, D. S. *et al.* Early arrival at the emergency department is associated with better collaterals, smaller established infarcts and better clinical outcomes with endovascular stroke therapy: SWIFT study. *J. Neurointerv. Surg.* http://dx.doi.org/10.1136/neurintsurg-2015-011758.
92. Ebinger, M. *et al.* Effect of the use of ambulance-based thrombolysis on time to thrombolysis in acute ischemic stroke: a randomized clinical trial. *JAMA* **311**, 1622–1631 (2014).
93. Hess, D. C. & Audebert, H. J. The history and future of telestroke. *Nat. Rev. Neurol.* **9**, 340–350 (2013).
94. Switzer, J. A. *et al.* Cost-effectiveness of hub-and-spoke telestroke networks for the management of acute ischemic stroke from the hospitals' perspectives. *Circ. Cardiovasc. Qual. Outcomes* **6**, 18–26 (2013).
95. Parker, S. A. *et al.* Establishing the first mobile stroke unit in the United States. *Stroke* **46**, 1384–1391 (2015).
96. Smith, W. S. *et al.* Mechanical thrombectomy for acute ischemic stroke: final results of the Multi MERCI trial. *Stroke* **39**, 1205–1212 (2008).
97. Samaniego, E. A., Dabus, G. & Linfante, I. Stenting in the treatment of acute ischemic stroke: literature review. *Front. Neurol.* **2**, 76 (2011).
98. Levy, E. I. *et al.* Midterm clinical and angiographic follow-up for the first Food and Drug Administration-approved prospective, Single-Arm Trial of Primary Stenting for Stroke: SARIS (stent-assisted recanalization for acute ischemic stroke). *Neurosurgery* **69**, 915–920 (2011).
99. John, S. *et al.* Initial experience using the 5MAX ACE reperfusion catheter in intra-arterial therapy for acute ischemic stroke. *J. Cerebrovasc. Endovasc. Neurosurg.* **16**, 350–357 (2014).

Author contributions

M.M. researched data for the article. All authors made substantial contributions to discussion of the content of the article, wrote the article, and reviewed and/or edited the manuscript before submission.

Competing interests statement

E.I.L. was a principal investigator of the Covidien US SWIFT PRIME trial, has had shareholder or ownership interests in Intratech Medical, CardinalHealth (formerly AccessClosure) and Blockade Medical, and has received compensation from Abbott for carotid stenting training for physicians. M.M. and H.R. declare no competing interests.

Review criteria

We searched the PubMed database for articles on the endovascular treatment of acute ischaemic stroke using the search terms "stroke", "endovascular", "thrombectomy", "thrombolysis", "stent retriever" and "intra-arterial" for all years. We reviewed full-length articles in English.

FURTHER INFORMATION

European Stroke Organization (ESO)-Karolinska Stroke Update: http://2014.strokeupdate.org/sites/default/files/Consensus_thrombectomy_ESO_Karolinska_ESMINT_ESNR_final.pdf

Royal College of Physicians Sentinel Stroke National Audit Programme (SSNAP) April–June 2015 summary report: https://www.strokeaudit.org/Documents/Results/National/AprJun2015/AprJun2015-SummaryReport.aspx

ALL LINKS ARE ACTIVE IN THE ONLINE PDF

© 2016 Macmillan Publishers Limited. All rights reserved